

EduVision_DV

Higher Education Performance Analytics Platform

Dashboard Storyboard

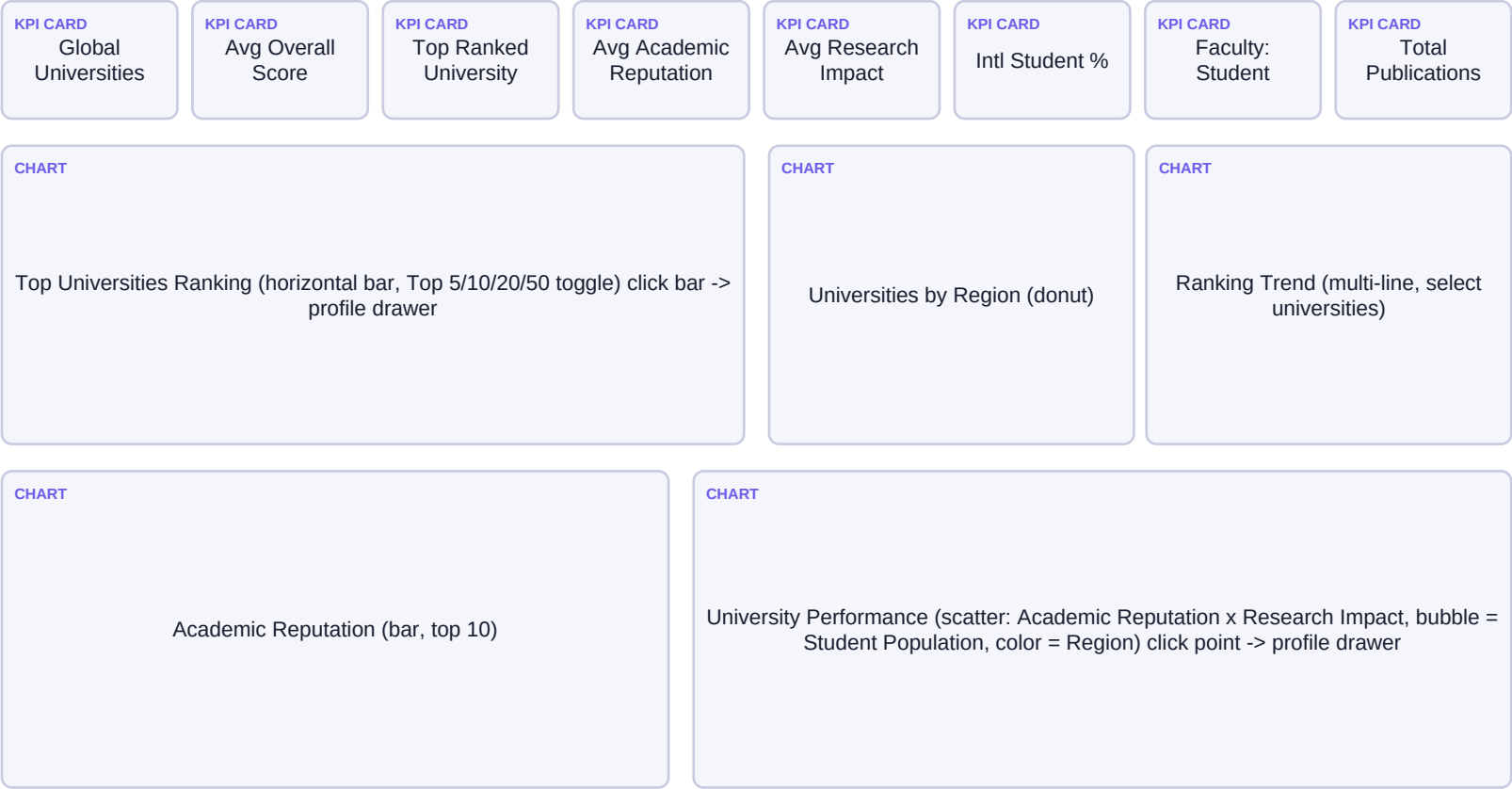
Milestone 2 · Module 4 — Dashboard Planning & Prototyping

- 1. University Overview — layout, filters, actions
- 2. Research Analytics — layout, filters, actions
- 3. Student Analytics — layout, filters, actions
- 4. Country Comparison — layout, filters, actions
- 5. Global Navigation & Filter Architecture
- 6. Dashboard Actions & Interactive Comparisons
- 7. Field Mapping — storyboard to university_final_dataset.xlsx

*Built against real, integrated data (7 sources, 1,373 universities, 4 ranking methodologies).
Every field named in this storyboard exists as a column in university_final_dataset.xlsx.*

FILTER

Global Filters (Year, Region, Country, Subject, Search)



Fields used

- global_ranking_score, academic_reputation_score, research_impact_score, international_student_pct, faculty_student_ratio, region, country, year

FILTER

Filters (Year, Region, Country, Subject) Page-local: University picker (for trend chart)

KPI CARD

Total Publications

KPI CARD

Total Citations

KPI CARD

Avg Citations/ Pub

KPI CARD

Research Productivity Idx

KPI CARD

Research Impact Score

KPI CARD

Top Research Institution

CHART

Publications by University (horizontal bar, top 10)

CHART

Citations by University (horizontal bar, top 10)

CHART

Research Productivity Trend (line, avg RPI across years)

CHART

Research Impact Comparison (scatter: Publications x Citations, bubble = Impact Score)

CHART

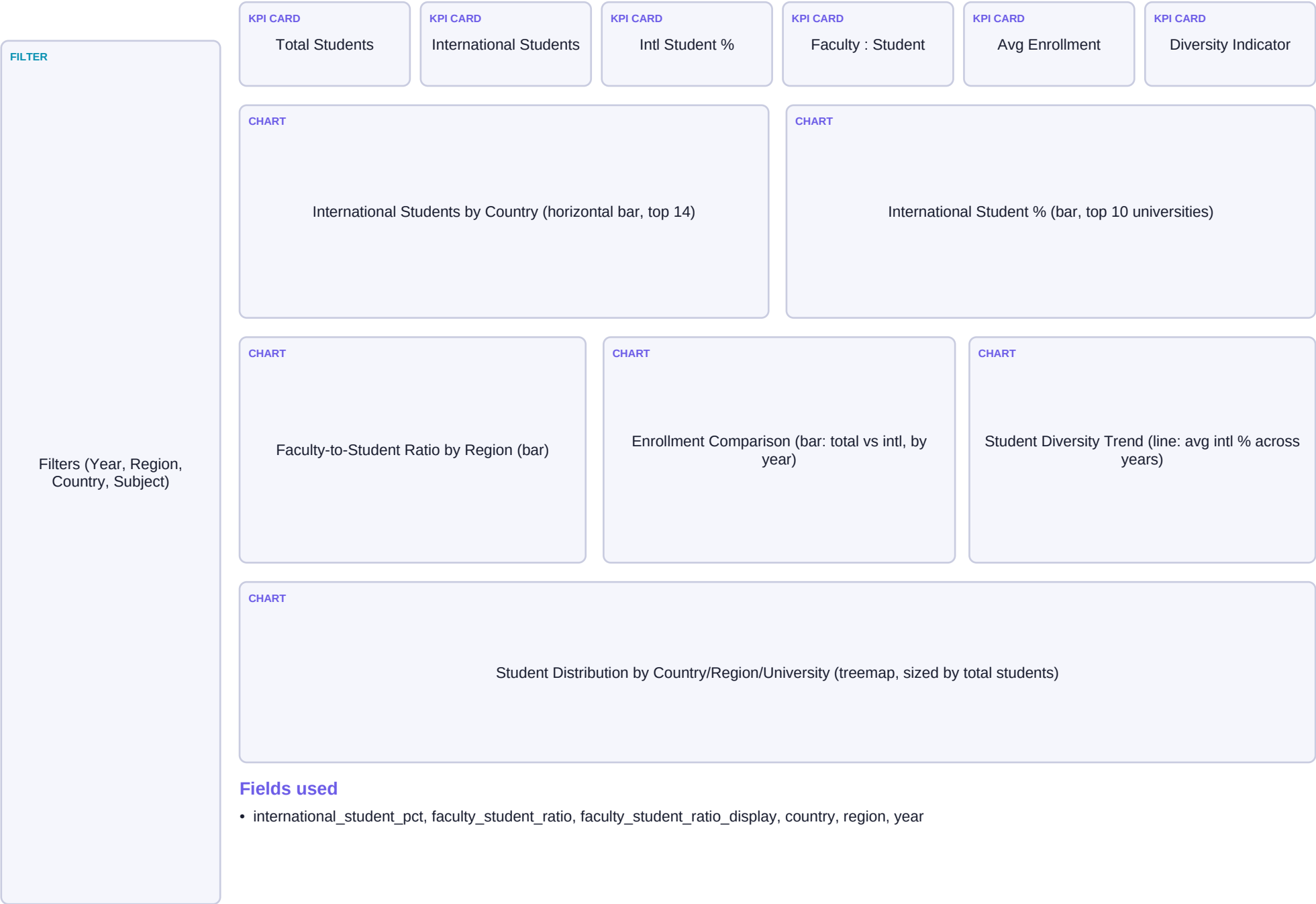
Research Trend by University (line: pubs/citations/impact for selected university)

TABLE

Top Research Institutions — ranked table (University, Country, Publications, Citations, Research Productivity Index, Research Impact Score) · sortable, exportable to CSV

Fields used

- research_productivity_index, research_impact_score, university_name, country, year



FILTER

Filters (Year, Region)
Page-local: Country multi-select (2-5) for comparison

KPI CARD

Countries

KPI CARD

Universities

KPI CARD

Avg Global Rank

KPI CARD

Avg Research Impact

KPI CARD

Avg Academic Reputation

KPI CARD

Intl Student %

MAP

World Distribution Map — schematic equirectangular plot, colored by selectable metric (Avg Ranking / Overall Score / Research Impact / Intl Student % / # Universities) circle size = universities · click country -> add to comparison

TABLE

Country Ranking — sortable table (Country, Universities, Avg Global Rank, Avg Score, Research Impact, Academic Reputation, Intl %)

CHART

Country Comparison — Radar chart (2-5 countries, Overall Score / Research Impact / Academic Reputation / Intl Student %)

CHART

Regional Analysis — grouped bar (universities count + avg score, by region)

Fields used

- country, region, global_ranking_score, research_impact_score, academic_reputation_score, international_student_pct

EduVision_DV — Global Navigation & Filter Architecture

Applies across all 4 dashboards

Navigation

- Left sidebar, collapsible: Overview / Research Analytics / Student Analytics / Country Comparison / University Comparison / Data Explorer / Methodology / University trend panel
- Top bar: logo, current dashboard name (breadcrumb), global filter bar, search, theme toggle
- Breadcrumb trail: EduVision_DV / <current dashboard> — always visible, one click back to Overview
- University name (anywhere it appears as a mark) is clickable -> opens the University Profile detail view as a modal/drawer, never a full navigation away from the current dashboard

Global Filters (persist across dashboards)

- Year — single-select, drives every KPI/chart on the active dashboard
- Region — single-select (All / North America / Europe / Asia / Oceania / South America / Africa)
- Country — single-select, options narrow automatically based on selected Region
- Subject Area — single-select (uses the university's classified subject field)
- Search — free text, filters university-level visuals + jumps to Profile on selection
- Reset Filters — one-click return to defaults (latest Year, All/All/All, empty search)
- Selected-filter chips shown under the filter bar so the active state is never hidden

Page-Local Filters (scoped to one dashboard)

- Country Comparison: Country multi-select (2-5) for the radar/comparison panel
- University Comparison: University multi-select (2-5) for the comparison table/radar
- Data Explorer: Sort field + direction, page size, in addition to the global filters
- Page-local filters reset when you navigate away — global filters do not

State & Persistence

- Filter state lives in one shared context object, read by every dashboard component
- Changing a global filter re-queries every visual on the CURRENT dashboard immediately
- Filter state is preserved when navigating between dashboards (Year stays selected, etc.)
- URL reflects active dashboard + key filters, so a link can be shared to a specific view

EduVision_DV — Dashboard Actions & Interactive Comparisons

Cross-filtering, drill-down, and comparison mechanics

Dashboard Actions (Tableau: Filter / Highlight / URL actions)

- FILTER action: clicking a Region on the donut (Overview) filters every other chart on that dashboard to that Region. Comparison: Region donut multi-select (2-5) sets some Overview
- FILTER action: clicking a Country on the World Map (Country Comparison) adds it to the 2-5 country Country Completion: used by the Radar chart (2-5), radar chart on global_ranking_score / research
- HIGHLIGHT action: hovering a university in the Ranking Trend legend highlights that same university's Parameter across the Top Universities on the World Map swaps the map's color encoding between Avg
- URL action: clicking a university name opens the University Profile sheet, filtered to that university_id — implemented as a Tableau dashboard-to-dashboard navigation action

Drill-down path

- Region (donut) -> Country (map/bar) -> University (profile)
- Each level's action clears/narrows the next level's filter — never both directions at once
- 'Reset Filters' always available at top-level to undo an accidental drill-down

Interactive Comparisons

- Every mark shows: university_name, country, the metric(s) plotted, and year on hover
- Bar/scatter marks additionally show num_sources and sources_covered, so a viewer can see wheth

Tooltips & hover detail

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EduVision_DV — Field Mapping

Storyboard fields -> university_final_dataset.xlsx columns

Storyboard field	Column in university_final_dataset.xlsx	Type	Used on
Global Ranking Score	<i>global_ranking_score</i>	Measure	Overview, Country
Research Impact Score	<i>research_impact_score</i>	Measure	Overview, Research, Country
Faculty-to-Student Ratio	<i>faculty_student_ratio / _display</i>	Measure / String	Overview, Student
International Student %	<i>international_student_pct</i>	Measure	Overview, Student, Country
Academic Reputation Score	<i>academic_reputation_score</i>	Measure	Overview, Country
Research Productivity Index	<i>research_productivity_index</i>	Measure	Research
University	<i>university_name / university_id</i>	Dimension	All
Country	<i>country</i>	Dimension	All
Region	<i>region</i>	Dimension	All
Year	<i>year</i>	Dim (discrete)	All
Source coverage	<i>num_sources / sources_covered</i>	Measure / String	Tooltips (all)

Notes for the Tableau build

- Load university_final_dataset.xlsx as a live or extracted connection — one flat sheet, one header row.
- Set 'year' to a discrete Dimension in Tableau immediately after connecting (it imports as a measure by default).
- Every measure column above has real NULLs where a source didn't publish that field for that university-year — use 'Special Values: Show as blank' rather than substituting a fabricated 0 or average.
- faculty_student_ratio_display ('1 : X') is a pre-formatted string for direct display in tooltips/labels; use the numeric faculty_student_ratio column for any actual sorting or aggregation.